6-DOF Quadrotor Simulation

Dynamics & INDI Control

Incremental Nonlinear Dynamic Inversion
Runge-Kutta 4 Integration · Quaternion Kinematics

Abstract. This document provides a self-contained mathematical derivation of a six-degree-of-freedom quadrotor simulation and its two-loop attitude controller based on Incremental Nonlinear Dynamic Inversion (INDI). We derive the rigid-body equations of motion in the body and world frames, introduce quaternion kinematics for singularity-free attitude representation, detail the propulsion model, and rigorously develop the INDI control law from first principles—contrasting it with classical Nonlinear Dynamic Inversion (NDI) and PID approaches. A quantitative assessment of the model's fidelity to physical reality is given, with explicit identification of omitted phenomena and their expected impact on simulation accuracy.

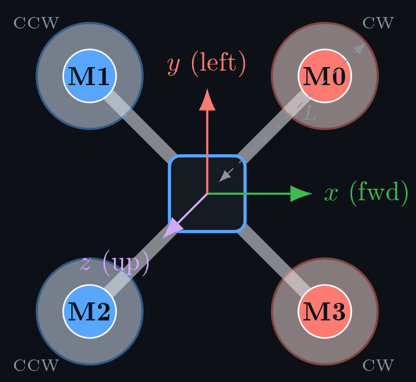
Technical Reference · Version 1.0

Contents

1 Vehicle Description and Coordinate Frames

1.1 Geometry and Motor Layout

The quadrotor is modelled as a rigid symmetric cross frame of arm length $L = 0.25\,\mathrm{m}$ with four rotors at the extremities.



Top view — red = CW (M0, M3), blue = CCW (M1, M2)

Figure 1: Quadrotor motor layout and body-frame axes.

1.2 Reference Frames

$\mathcal{W}$ World (inertial) frame: right-handed, z pointing up (ENU convention).

$\mathcal{B}$ Body frame: origin at centre of mass, x forward, y left, z up. Attached to the airframe.

$\mathbf{R}$ Rotation matrix $\mathbf{R} \in \mathrm{SO}(3)$ transforming vectors from $\mathcal{B}$ to $\mathcal{W}$: $\boldsymbol{v}_{\mathcal{W}} = \mathbf{R}\,\boldsymbol{v}_{\mathcal{B}}$.

2 Rigid-Body Dynamics

2.1 Translational Equations of Motion

Applying Newton's second law in $\mathcal{W}$:

$$m\,\dot{\boldsymbol{v}} = \mathbf{R}\,\boldsymbol{F}_{\mathcal{B}} + \boldsymbol{F}_{\mathrm{drag}} + \boldsymbol{F}_g \tag{1}$$

where $\boldsymbol{F}_{\mathcal{B}} = [0,\ 0,\ T_{\mathrm{tot}}]^\top$ is the total thrust in $\mathcal{B}$ (rotors thrust along the body z-axis), $\boldsymbol{F}_g = mg\,\hat{e}_z^-$ is gravity, and the aerodynamic drag is:

$$\boldsymbol{F}_{\mathrm{drag}} = -\tfrac{1}{2}\,\rho\,C_D\,A\,\|\boldsymbol{v}_{\mathrm{rel}}\|\,\boldsymbol{v}_{\mathrm{rel}}, \qquad \boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v} - \boldsymbol{w}_{\mathrm{wind}} \tag{2}$$

with $\rho = 1.225\,\mathrm{kg\,m^{-3}}$, $C_D = 1.0$, $A = 0.05\,\mathrm{m}^2$.

2.2 Rotational Equations of Motion — Euler's Equation

Let $\boldsymbol{\omega} = [p,\ q,\ r]^\top$ be the angular velocity expressed in $\mathcal{B}$. Euler's equation for a rigid body with inertia tensor $\mathbf{I}$ is:

$$\mathbf{I}\,\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\text{total}} - \boldsymbol{\omega} \times (\mathbf{I}\,\boldsymbol{\omega}) \tag{3}$$

The term $\boldsymbol{\omega} \times (\mathbf{I}\,\boldsymbol{\omega})$ is the *gyroscopic* coupling inherent to a rotating rigid body; it vanishes only when $\boldsymbol{\omega}$ is aligned with a principal axis. For our diagonal inertia tensor $\mathbf{I} = \operatorname{diag}(I_{xx}, I_{yy}, I_{zz})$ this expands to:

$$\boldsymbol{\omega} \times (\mathbf{I}\,\boldsymbol{\omega}) = \begin{bmatrix} (I_{zz} - I_{yy})\,q\,r \\ (I_{xx} - I_{zz})\,p\,r \\ (I_{yy} - I_{xx})\,p\,q \end{bmatrix}. \tag{4}$$

Inertia values. $I_{xx} = I_{yy} = 0.02\,\mathrm{kg\,m^2}$, $I_{zz} = 0.04\,\mathrm{kg\,m^2}$. The higher I_{zz} reflects the mass distribution in the horizontal plane.

2.3 Total Body Torque Decomposition

$$\boldsymbol{\tau}_{\text{total}} = \underbrace{\boldsymbol{\tau}_{\text{thrust}}}_{\text{motor forces}} + \underbrace{\boldsymbol{\tau}_{\text{gyro}}}_{\text{rotor angular momentum}} + \underbrace{\boldsymbol{\tau}_{\text{damp}}}_{\text{linear damping}} \tag{5}$$

2.3.1 Thrust-Induced Torques

Motor i produces thrust $T_i = k_T\,\Omega_i^2$ and reaction torque $Q_i = k_Q\,\Omega_i^2$. The sign convention of motor positions (see fig. 1) gives:

$$\tau_x = L\,(T_3 - T_1) \tag{6}$$

$$\tau_y = L\,(T_0 - T_2) \tag{7}$$

$$\tau_z = Q_0 - Q_1 + Q_2 - Q_3 = \frac{k_Q}{k_T}(T_0 - T_1 + T_2 - T_3) \tag{8}$$

Motors M1 and M3 are on the $\pm y$ arms. Increasing T_1 rolls the vehicle negative (right), while increasing T_3 rolls it positive (left), hence $\tau_x = L(T_3 - T_1)$. This must be consistent with the control allocation matrix $\mathbf{A}$ in section 8.

2.3.2 Gyroscopic Torque from Rotors

Each spinning rotor carries angular momentum $h_i = J_r\,\Omega_i$ along $\hat{e}_z$. The net rotor angular momentum and the resulting gyroscopic body torque are:

$$\boldsymbol{H}_r = J_r \sum_{i=0}^{3} s_i\,\Omega_i\,\hat{e}_z = J_r\,\Gamma_r\,\hat{e}_z, \quad s_i \in \{+1, -1\} \tag{9}$$

$$\boldsymbol{\tau}_{\text{gyro}} = \boldsymbol{\omega} \times \boldsymbol{H}_r \tag{10}$$

where $\boldsymbol{s} = [+1, -1, +1, -1]$ and $J_r = 1 \times 10^{-5}\,\mathrm{kg\,m^2}$. At hover all four motors spin at Ω_{hov}, so $\Gamma_r = \Omega_{\mathrm{hov}}(1 - 1 + 1 - 1) = 0$ and the gyroscopic torque vanishes identically. During aggressive manoeuvres the differential speed is non-zero and this term becomes significant, coupling pitch and roll.

2.3.3 Viscous Angular Damping

$$\boldsymbol{\tau}_{\mathrm{damp}} = -k_\omega\,\boldsymbol{\omega}, \quad k_\omega = 0.02\ \mathrm{N\,m\,s\,rad^{-1}} \tag{11}$$

This models aggregate structural and aerodynamic damping of body rotation.

3 Propulsion Model

3.1 Motor First-Order Lag

Real brushless DC motors cannot respond instantaneously; electrical inductance and mechanical inertia impose a lag. We model this as a first-order system with time constant τ_m:

$$\dot{\Omega}_i = \frac{\Omega_i^{\mathrm{cmd}} - \Omega_i}{\tau_m}, \quad \tau_m = 0.02\,\mathrm{s} \tag{12}$$

The bandwidth of this model is $f_{\mathrm{bw}} = (2\pi\tau_m)^{-1} \approx 8\,\mathrm{Hz}$, consistent with typical racing-class ESC/motor combinations.

3.2 Aerodynamic Thrust and Torque Laws

In the blade-element / momentum theory limit, steady-state thrust and reaction torque are quadratic in rotor speed:

$$T_i = k_T\,\Omega_i^2, \quad k_T = 10^{-5}\ \mathrm{N\,(rad/s^{-2})} \tag{13}$$

$$Q_i = k_Q\,\Omega_i^2, \quad k_Q = 2 \times 10^{-6}\ \mathrm{N\,m\,(rad/s^{-2})} \tag{14}$$

3.3 Hover Equilibrium

At hover, total thrust equals weight and torques balance. By symmetry, all four motors run at the same speed Ω_{hov}:

$$4\,k_T\,\Omega_{\mathrm{hov}}^2 = mg \implies \Omega_{\mathrm{hov}} = \sqrt{\frac{mg}{4\,k_T}} = \sqrt{\frac{1.0 \times 9.81}{4 \times 10^{-5}}} \approx 495\,\mathrm{rad\,s^{-1}} \quad (\approx 4727\,RPM) \tag{15}$$

3.4 Ground Effect

Within two rotor radii of the ground, the wake recirculation increases effective thrust. We use the Cheeseman–Bennett correction:

$$T_i^{\text{eff}} = \frac{T_i}{1 - \left(\dfrac{R}{4z}\right)^2}, \quad 0 < z < 2R, \quad R = 0.1\,\text{m} \tag{16}$$

4 Quaternion Kinematics

4.1 Motivation

Euler angles suffer from *gimbal lock* at $\theta = \pm\pi/2$: a degree of freedom is lost and the parameterisation becomes singular. Rotation matrices $\mathbf{R} \in \mathrm{SO}(3)$ avoid this but carry 9 elements with 6 constraints. **Unit quaternions** $\bar{\boldsymbol{q}} \in S^3$ offer the best compromise: 4 elements, 1 constraint (unit norm), no singularities, and efficient composition:

$$\bar{\boldsymbol{q}} = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} \in \mathbb{R}^4, \quad \|\bar{\boldsymbol{q}}\| = 1. \tag{17}$$

4.2 Rotation Matrix from Quaternion

$$\mathbf{R} = \begin{pmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_z q_w) & 2(q_x q_z + q_y q_w) \\ 2(q_x q_y + q_z q_w) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_x q_w) \\ 2(q_x q_z - q_y q_w) & 2(q_y q_z + q_x q_w) & 1 - 2(q_x^2 + q_y^2) \end{pmatrix} \tag{18}$$

4.3 Quaternion Kinematic Equation

The quaternion evolves according to:

$$\dot{\bar{\boldsymbol{q}}} = \frac{1}{2}\,\boldsymbol{\Xi}(\boldsymbol{\omega})\,\bar{\boldsymbol{q}} \tag{19}$$

where the skew-symmetric matrix $\boldsymbol{\Xi}$ is:

$$\boldsymbol{\Xi}(\boldsymbol{\omega}) = \begin{pmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{pmatrix}, \quad \boldsymbol{\omega} = [p,\, q,\, r]^\top. \tag{20}$$

4.4 Euler Angle Extraction from Quaternion

For display and the outer control loop, intrinsic ZYX Euler angles are recovered analytically:

$$\phi = \arctan\frac{2(q_w q_x + q_y q_z)}{1 - 2(q_x^2 + q_y^2)} \tag{21}$$

$$\theta = \arcsin\operatorname{clip}\big(2(q_w q_y - q_z q_x),\, -1,\, 1\big) \tag{22}$$

$$\psi = \arctan\frac{2(q_w q_z + q_x q_y)}{1 - 2(q_y^2 + q_z^2)} \tag{23}$$

4.5 Quaternion Re-normalisation

Numerical integration accumulates floating-point error that slowly destroys the unit-norm constraint. After each RK4 step we renormalise:

$$\bar{\boldsymbol{q}} \leftarrow \frac{\bar{\boldsymbol{q}}}{\|\bar{\boldsymbol{q}}\|}. \tag{24}$$

First-order renormalisation methods (e.g. $\bar{\boldsymbol{q}} \leftarrow \frac{1}{2}(3 - \|\bar{\boldsymbol{q}}\|^2)\bar{\boldsymbol{q}}$) are slightly faster but the direct division is trivially cheap at 200 Hz.

5 Numerical Integration: Runge-Kutta 4

5.1 State Vector

The complete simulation state is:

$$\boldsymbol{x} = \big[\underbrace{\boldsymbol{p}^\top}_{3},\ \underbrace{\boldsymbol{v}^\top}_{3},\ \underbrace{\bar{\boldsymbol{q}}^\top}_{4},\ \underbrace{\boldsymbol{\omega}^\top}_{3},\ \underbrace{\boldsymbol{\Omega}^\top}_{4}\big]^\top \in \mathbb{R}^{17} \tag{25}$$

Index	Symbol	Dim	Description	Units
0:3	$\boldsymbol{p}$	3	World-frame position	m
3:6	$\boldsymbol{v}$	3	World-frame velocity	$\mathrm{m\,s^{-1}}$
6:10	$\bar{\boldsymbol{q}}$	4	Unit quaternion	–
10:13	$\boldsymbol{\omega}$	3	Body angular rate	$\mathrm{rad\,s^{-1}}$
13:17	$\boldsymbol{\Omega}$	4	Motor speeds	$\mathrm{rad\,s^{-1}}$

5.2 RK4 Scheme

The ODE $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x})$ is integrated at fixed step $\Delta t = 5\,\mathrm{ms}$ (200 Hz):

$$\boldsymbol{k}_1 = \boldsymbol{f}(\boldsymbol{x}_k) \tag{26}$$

$$\boldsymbol{k}_2 = \boldsymbol{f}\big(\boldsymbol{x}_k + \tfrac{\Delta t}{2}\boldsymbol{k}_1\big) \tag{27}$$

$$\boldsymbol{k}_3 = \boldsymbol{f}\big(\boldsymbol{x}_k + \tfrac{\Delta t}{2}\boldsymbol{k}_2\big) \tag{28}$$

$$\boldsymbol{k}_4 = \boldsymbol{f}\big(\boldsymbol{x}_k + \Delta t\,\boldsymbol{k}_3\big) \tag{29}$$

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \frac{\Delta t}{6}\left(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\right) \tag{30}$$

RK4 achieves $\mathcal{O}(\Delta t^4)$ local truncation error. The stiffest mode in our system is the motor lag with time constant $\tau_m = 20\,\text{ms}$; the stability ratio $\Delta t/\tau_m = 0.25$ is well within the RK4 stability region (< 2.8 for real eigenvalues).

6 Why INDI? Control Design Motivation

6.1 Classical Control Limitations

A naïve PID on Euler angles is simple but exhibits well-known limitations for a 6-DOF vehicle:

- **Euler coupling:** The Euler angle rates $(\dot{\phi}, \dot{\theta}, \dot{\psi})$ are not body rates ($\boldsymbol{\omega}$); they are related by a nonlinear kinematic transformation that becomes singular near $\theta = \pm 90°$.
- **Model sensitivity:** Cross-coupling terms $\boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})$ and gyroscopic rotor torques must be explicitly cancelled or treated as disturbances, degrading performance.
- **Fixed gains:** Linear gains designed at one operating point degrade as the vehicle manoeuvres aggressively.

6.2 Classical NDI and Its Model-Dependency Problem

Nonlinear Dynamic Inversion (NDI) solves the coupling problem exactly by directly inverting the nonlinear plant. For the rotational dynamics:

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}[\boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})] =: f(\boldsymbol{\omega}) + g \cdot \boldsymbol{\tau} \tag{31}$$

NDI inverts this to command torques that produce any desired $\dot{\boldsymbol{\omega}}_{\text{des}}$:

$$\boldsymbol{\tau}_{\text{NDI}} = \mathbf{I}\,\dot{\boldsymbol{\omega}}_{\text{des}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) - \boldsymbol{\tau}_{\text{gyro}} - \boldsymbol{\tau}_{\text{damp}} \tag{32}$$

Critical flaw. eq. (32) requires an accurate model of every force and torque: gyroscopic coupling, aerodynamic damping, rotor torques, ground effect. Any model error $\delta\boldsymbol{\tau}$ directly becomes a tracking error that the outer loop must compensate.

6.3 INDI: Sensor-Based Incremental Inversion

INDI replaces the model terms in eq. (32) with a real-time *measurement*. Starting from the true dynamics at time t_k:

$$\dot{\boldsymbol{\omega}}(t_k) = \mathbf{I}^{-1}[\boldsymbol{\tau}(t_k) - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) + \text{all disturbances}] \tag{33}$$

we *measure* $\dot{\boldsymbol{\omega}}_{\text{meas}} \approx (\boldsymbol{\omega}_k - \boldsymbol{\omega}_{k-1})/\Delta t$ and assume that within one timestep the non-control terms change negligibly. To shift the angular acceleration from its current value to a desired value $\dot{\boldsymbol{\omega}}_{\text{des}}$, we need only add a torque increment:

$$\boxed{\delta\boldsymbol{\tau} = \mathbf{I}(\dot{\boldsymbol{\omega}}_{\text{des}} - \dot{\boldsymbol{\omega}}_{\text{meas}})} \tag{34}$$

The full commanded torque is then updated *incrementally*:

$$\boldsymbol{\tau}_{\text{cmd}}[k] = \boldsymbol{\tau}_{\text{cmd}}[k-1] + \delta\boldsymbol{\tau} \tag{35}$$

All disturbances (gyroscopic coupling, aerodynamic damping, ground effect, rotor gyroscopic effects) are *already present* in the current $\dot{\boldsymbol{\omega}}_{\text{meas}}$. By commanding only the *difference* $\dot{\boldsymbol{\omega}}_{\text{des}} - \dot{\boldsymbol{\omega}}_{\text{meas}}$, we cancel them automatically, without needing to identify or model them. This makes INDI inherently robust to model uncertainty and unmodelled disturbances, including wind gusts and changing payload.

6.4 Stability Analysis

Let the attitude tracking error be $e_\phi = \phi - \phi_{\text{cmd}}$. The closed-loop system under INDI with proportional rate gain k_r and attitude gain k_a is (linearised about hover):

$$\ddot{e}_\phi + k_r\,\dot{e}_\phi + k_r\,k_a\,e_\phi = 0 \tag{36}$$

This is a standard second-order system with natural frequency $\omega_n = \sqrt{k_r k_a}$ and damping ratio $\zeta = k_r/(2\omega_n) = \frac{1}{2}\sqrt{k_r/k_a}$.

With $k_r = 8.0$ and $k_a = 2.0$: $\omega_n = \sqrt{16} = 4\,\text{rad}\,\text{s}^{-1}$ and $\zeta = 8/(2 \cdot 4) = 1.0$ (critically damped) — providing fast, non-oscillatory attitude response.

7 Two-Loop INDI Architecture

7.1 Architecture Overview

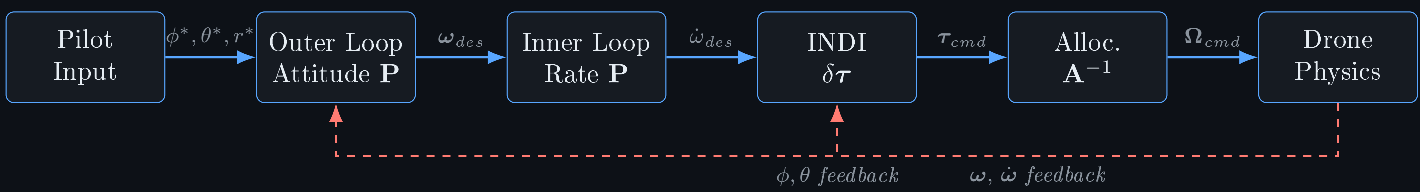


Figure 2: Two-loop INDI control architecture. Dashed red = sensor feedback.

7.2 Outer Loop: Attitude Controller

The outer loop converts attitude errors to body-rate commands. Attitude errors are computed from the quaternion-extracted Euler angles:

$$\boldsymbol{\omega}_{\text{des}} = \mathbf{K}_a \begin{bmatrix} \text{clip}(\phi^* - \phi,\ -0.5,\ 0.5) \\ \text{clip}(\theta^* - \theta,\ -0.5,\ 0.5) \\ r^* \end{bmatrix}, \quad \mathbf{K}_a = \text{diag}(2.0,\ 2.0,\ 1.0) \tag{37}$$

The clipping prevents the controller from commanding unrealisable roll/pitch rates during large step commands. Yaw is passed through as a direct rate command r^*.

7.3 Inner Loop: Rate Controller

The rate error feeds into a proportional gain to produce the desired angular acceleration:

$$\dot{\boldsymbol{\omega}}_{\text{des}} = \mathbf{K}_r \;\text{clip}\big(\boldsymbol{\omega}_{\text{des}} - \boldsymbol{\omega},\; -5,\; 5\big), \quad \mathbf{K}_r = \text{diag}(8.0,\; 8.0,\; 3.0) \tag{38}$$

The clipping saturates the rate error to prevent integrator-like wind-up in the INDI increment accumulation.

7.4 INDI Step and Torque Increment

$$\dot{\boldsymbol{\omega}}_{\text{meas}}[k] = \frac{\boldsymbol{\omega}[k] - \boldsymbol{\omega}[k-1]}{\Delta t} \tag{39}$$

$$\delta\boldsymbol{\tau}[k] = \mathbf{I}(\dot{\boldsymbol{\omega}}_{\text{des}}[k] - \dot{\boldsymbol{\omega}}_{\text{meas}}[k]) \tag{40}$$

$$\boldsymbol{\tau}_{\text{cmd}}[k] = \boldsymbol{\tau}_{\text{cmd}}[k-1] + \text{clip}(\delta\boldsymbol{\tau}[k],\; -0.5,\; 0.5) \tag{41}$$

After saturation rescaling (see section 8), the stored $\boldsymbol{\tau}_{\text{cmd}}[k-1]$ must reflect the *actually applied* torque, not the pre-saturation value. Otherwise the INDI baseline drifts and the increment equation eq. (41) becomes unstable.

8 Control Allocation

8.1 Allocation Matrix

The virtual control vector $\boldsymbol{u} = [T_{\text{tot}},\; \tau_x,\; \tau_y,\; \tau_z]^\top$ is mapped to individual motor thrusts $\boldsymbol{T} = [T_0, T_1, T_2, T_3]^\top$:

$$\underbrace{\begin{bmatrix} T_{\text{tot}} \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix}}_{\boldsymbol{u}} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -L & 0 & L \\ L & 0 & -L & 0 \\ \kappa & -\kappa & \kappa & -\kappa \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} T_0 \\ T_1 \\ T_2 \\ T_3 \end{bmatrix}}_{\boldsymbol{T}}, \quad \kappa = \frac{k_Q}{k_T} = 0.2 \tag{42}$$

Motor thrusts are obtained by matrix inversion: $\boldsymbol{T} = \mathbf{A}^{-1}\boldsymbol{u}$.

8.2 Saturation and Thrust Priority

Physical motor thrusts are bounded: $T_{\min} = 0.2\,T_{\text{hov}}$ to $T_{\max} = 3.0\,T_{\text{hov}}$. When $\mathbf{A}^{-1}\boldsymbol{u}$ violates these bounds, a priority scheme preserves total thrust while reducing torque demands:

$$\boldsymbol{u}_{\text{scaled}} = \begin{bmatrix} T_{\text{tot}} \\ \sigma\,\tau_x \\ \sigma\,\tau_y \\ \sigma\,\tau_z \end{bmatrix}, \quad \sigma \in \{1.0,\; 0.5,\; 0.2,\; 0.0\} \tag{43}$$

Choose the largest σ such that $\mathbf{A}^{-1}\boldsymbol{u}_{\text{scaled}} \in [T_{\min}, T_{\max}]^4$. If no σ works: $T_i = T_{\text{tot}}/4$ (pure thrust, zero torque).

8.3 Motor Speed Commands

$$\Omega_i^{\text{cmd}} = \sqrt{\max(T_i,\, 0) \,/\, k_T} \tag{44}$$

9 Altitude Hold

A proportional-derivative controller maintains the pilot-commanded altitude z^*:

$$T_{\text{cmd}} = mg + K_{p,z}(z^* - z) + K_{d,z}(-\dot{z}), \quad K_{p,z} = 12.0, \quad K_{d,z} = 4.0 \tag{45}$$

Thrust is clipped to $[0.25\,mg,\ 1.85\,mg]$. The altitude setpoint z^* is adjusted by the operator in $0.5\,\text{m}$ steps within $[0.3,\ 8.0]\,\text{m}$.

10 Physical Fidelity: What We Capture and What We Omit

10.1 Modelled Physics

1. **6-DOF rigid-body dynamics** — full Newton–Euler equations with inertia coupling $\boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})$.
2. **Quaternion kinematics** — singularity-free, globally valid attitude representation.
3. **Motor first-order lag** ($\tau_m = 20\,\text{ms}$) — limits bandwidth, prevents step commands from being instantaneously realised.
4. **Quadratic thrust and torque** ($T \propto \Omega^2$, $Q \propto \Omega^2$) — correct aerodynamic scaling.
5. **Rotor gyroscopic torque** — couples pitch and roll during differential speed manoeuvres.
6. **Rigid-body gyroscopic term** $\boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})$ — inherent in Euler's equation, couples all three body rates.
7. **Quadratic aerodynamic drag** — velocity-squared body drag, correct for low-Reynolds turbulent flows.
8. **Cheeseman–Bennett ground effect** — thrust augmentation near the ground.
9. **Wind disturbance** — additive wind vector on relative velocity (can be non-zero).
10. **Quaternion renormalisation** — prevents SO(3) constraint drift.

10.2 Omissions and Their Expected Impact

1. **Blade flapping and lead-lag dynamics**
Real rotors are flexible; blades flap (out-of-plane) and lead/lag (in-plane) under aerodynamic loading, introducing first-order roll/pitch moments that couple with translational velocity. *Impact: moderate at low airspeed; significant above* $\sim 5\,\mathrm{m\,s^{-1}}$.

2. **Translational drag on rotors (H-force)**
When the drone moves horizontally, the rotor disc experiences an asymmetric inflow that produces a net in-plane force (*H-force*) and a pitching moment. This creates a "pitch-back" tendency in forward flight not captured by our body-frame drag model. *Impact: low at hover, grows as* v^2.

3. **Induced velocity dynamics**
Momentum theory gives a static inflow velocity $v_i = \sqrt{T/2\rho A}$; in reality v_i has its own transient dynamics (dynamic inflow). When thrust changes rapidly the actual induced velocity lags, causing a transient thrust overshoot. *Impact: transient bandwidth errors on the order of the inflow time constant* $\sim 10\,\mathrm{ms}$.

4. **Motor current and voltage dynamics / battery sag**
Our model assumes unlimited electrical power. In reality, battery internal resistance causes voltage sag under high current, reducing maximum achievable Ω. *Impact: saturation headroom is overestimated; negligible for qualitative control studies.*

5. **Structural flexibility**
Arm and frame vibrations introduce high-frequency dynamics ($f > 100\,\mathrm{Hz}$) not relevant to our 200 Hz simulation but important for vibration-sensitive sensors. *Impact: negligible below* 100 Hz.

6. **Propeller inflow interference**
Adjacent rotors' wakes interact, reducing effective thrust. Particularly significant at small arm lengths relative to rotor diameter. *Impact: a constant multiplicative bias on* k_T*; calibrated out in practice.*

7. **Non-linear propeller stall and vortex ring state**
At high descent rates the rotor operates in vortex ring state, where thrust collapses unpredictably. *Impact: catastrophic for aggressive descents; not present in our model.*

8. **Sensor noise, latency, and quantisation**
Our INDI law uses $\dot{\boldsymbol{\omega}}_{\mathrm{meas}}$ computed from a clean $\boldsymbol{\omega}$ signal. Real gyroscopes add noise (amplified by differentiation), bias, and temperature drift. *Impact: major; in practice requires a low-pass filter on* $\dot{\boldsymbol{\omega}}_{meas}$ *that introduces phase lag and reduces INDI bandwidth.*

9. **Motor speed measurement delay**
We assume $\boldsymbol{\Omega}$ is immediately known. Real ESCs report speed with latency ($\sim 1\,\mathrm{ms}$), introducing a sample-and-hold delay in the allocation chain. *Impact: minor at* 200 Hz.

10. **Variable air density / wind turbulence spectrum**
We use a constant ρ and steady wind. Real outdoor flights encounter turbulence with a Dryden or von Kármán power spectrum. *Impact: attitude disturbances; INDI partially rejects these via its disturbance-cancellation property.*

10.3 Fidelity Summary Table

Effect	Modelled	Omitted	Impact
Rigid-body inertia coupling	✓		–
Quaternion singularity-free att	✓		–
Motor lag τ_m	✓		–
Rotor gyroscopic torque	✓		–
Ground effect	✓		–
Body aerodynamic drag	✓		–
Blade flapping / H-force		×	Moderate
Dynamic inflow		×	Low–Med
Battery sag		×	Low
Sensor noise / gyro latency		×	High
Vortex ring state		×	High
Propeller interference		×	Low
Wind turbulence spectrum		×	Medium

11 Summary

- **Model-free disturbance rejection:** eq. (34) cancels all disturbances captured in $\dot{\boldsymbol{\omega}}_{\text{meas}}$ without explicit identification.
- **Incremental safety:** Only small $\delta\boldsymbol{\tau}$ are added each step; the system cannot command a sudden large torque spike from numerical error.
- **Critically damped response:** The gain choices give $\zeta = 1.0$, $\omega_n = 4\,\text{rad}\,\text{s}^{-1}$ — fast attitude settling without overshoot.
- **Thrust priority saturation:** Altitude hold is never sacrificed for attitude correction.
- **Quaternion robustness:** No gimbal lock; valid for aerobatic manoeuvres and inverted flight.
- **RK4 accuracy:** $\mathcal{O}(\Delta t^4) = \mathcal{O}(6.25 \times 10^{-13})$ local truncation error at 200 Hz — negligible compared to omitted physical effects.

All equations are implemented verbatim in `dronedynamics.py` and `control.py`.